No non-zero rest mass for a Dirac field across future conformal infinity: a spin-$\frac{1}{2}$ counterpart of Friedrich's scalar condition and its consequence for the conformal crossover of Conformal Cyclic Cosmology

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14 October 2026 (version 1, working paper)

Abstract

In Penrose's Conformal Cyclic Cosmology the future conformal boundary $\mathscr{I}^+$ of one aeon is identified with the initial singularity of the next, and the matter fields are required to extend smoothly, after conformal rescaling, across that surface. Friedrich showed that a massive scalar field can do so only if its mass satisfies $m^2 = 2\lambda/3$, which on de Sitter space is exactly the value at which the minimally coupled field becomes indistinguishable from the conformally coupled massless one. We examine the same question for a Dirac field on an asymptotically de Sitter background. The conformally rescaled Dirac equation carries the mass as m/Ω, and no dimensionless curvature coupling can play the role that $R/6$ plays for the scalar. We show that for every mode and every $m \neq 0$ the rescaled field keeps a constant norm but acquires a phase $\exp(\pm \mathrm{i}(m/H)\ln\Omega)$ that winds without bound as $\Omega \to 0$: the rescaled field has no limit at $\mathscr{I}^+$, not merely a non-smooth one. For the electron, with $m_e/H_\infty \simeq 4 \times 10^{38}$, the phase turns by that many radians per e-fold of the conformal factor. The consequence is that a conformal crossover which is even continuous for a Dirac field is possible only if the physical mass of every Dirac field present in the state vanishes at $\mathscr{I}^+$. Penrose's requirement that rest mass fade near the crossover is therefore not a convenience but a necessity, and it is not met by the Standard Model, in which the electron is stable and its mass constant. We state precisely what is and is not established, and the alternatives that remain open.

1 Introduction

Conformal Cyclic Cosmology (CCC) [1, 2, 3] proposes that the future conformal boundary $\mathscr{I}^+$ of a universe with positive cosmological constant be identified, through a conformal rescaling, with the initial singularity of a subsequent universe (the next "aeon"). The construction rests on two geometric facts: the metric of an asymptotically de Sitter spacetime, multiplied by a conformal factor Ω^2 with $\Omega \to 0$ at late times, extends regularly to a spacelike hypersurface $\mathscr{I}^+$ at finite conformal distance; and an isotropic initial singularity, multiplied by a conformal factor Ω^{-2}, likewise extends regularly to a spacelike hypersurface. CCC identifies the two surfaces and requires that the matter content, suitably rescaled, pass from one side to the other.

This requirement has a well-known price. The rescaling is a Weyl transformation, and massless, conformally invariant fields (Maxwell, the conformally coupled scalar, the massless Dirac field) are the only ones that behave well under it. Penrose therefore postulates that near $\mathscr{I}^+$ all rest mass has become negligible, or has faded away, so that the matter content

is effectively conformally invariant [1]. Whether this can be true for the Standard Model, in which the electron is stable and its mass is a constant of nature, is the question that this paper makes precise for the Dirac field.

For a scalar field the question was settled by Friedrich [4]. Working with the conformal field equations coupled to a massive scalar, he proved that the coupled system extends smoothly across $\mathscr{I}^+$ if and only if the mass takes the particular value $m^2 = 2\lambda/3$, where λ is the cosmological constant; otherwise terms proportional to $(m^2 - 2\lambda/3)\,\Omega^{-k}$, $k = 1, 2$, obstruct the extension. He also noted that no known process would justify a mass taking that value, or fading to it. We make two remarks on this result in Section 3: on de Sitter space, $2\lambda/3 = R/6$, so the admissible mass is exactly the one at which a minimally coupled scalar coincides with the conformally coupled massless scalar; and the obstruction for a generic mass is an obstruction to *smoothness*, the rescaled field itself remaining bounded.

For a Dirac field there is no analogous condition. The reason is elementary once the equations are written in the conformal frame (Section 4): the massless Dirac operator is conformally covariant, the mass term becomes m/Ω, and there is no local, dimensionless curvature coupling that could cancel it, because every local curvature scalar has even dimension while m has dimension one. What replaces Friedrich's condition is a sharper obstruction. We show (Proposition 1) that for every mode, every non-zero mass and every state, the rescaled Dirac field keeps a constant norm but has no limit at $\mathscr{I}^+$, because its phase winds as $\exp(\pm \mathrm{i}(m/H)\ln\Omega)$. The obstruction for a spinor is thus already an obstruction to *continuity*. In Section 5 we draw the consequence for CCC: a crossover that is continuous for Dirac fields exists only if the physical mass of every Dirac field present in the state vanishes at $\mathscr{I}^+$; with the Standard-Model electron this is not the case. Section 6 states what the argument does not establish and which alternatives remain.

The argument is short, and most of its ingredients are standard. What is new, to our knowledge, is the explicit statement that the spinor obstruction is to continuity rather than smoothness, that it holds mode by mode independently of the state, and that it cannot be removed by any curvature coupling of the kind that rescues the scalar. The paper is written to be checked, and we would welcome a counter-argument.

2 Setting and conventions

Throughout, g denotes the physical metric, of signature $(-,+,+,+)$, and $\hat{g} = \Omega^2 g$ the unphysical (conformally rescaled) metric, with $\Omega > 0$ in the physical region and $\Omega = 0$ on $\mathscr{I}^+$. This is the convention of Penrose; in Friedrich [4] the roles of hatted and unhatted metrics are interchanged.

We work on the expanding patch of de Sitter space with Hubble constant H, in spatially flat slicing,

$$g = -\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\mathbf{x}^2, \qquad a = e^{Ht}, \qquad \eta = \int \frac{\mathrm{d}t}{a} = -\frac{1}{aH} \in (-\infty, 0). \tag{1}$$

In conformal time $g = a(\eta)^2\,(-\mathrm{d}\eta^2 + \mathrm{d}\mathbf{x}^2)$, so that the natural choice $\Omega = 1/a = -H\eta$ gives a *flat* unphysical metric $\hat{g} = -\mathrm{d}\eta^2 + \mathrm{d}\mathbf{x}^2$, and $\mathscr{I}^+$ is the hypersurface $\eta = 0$, reached at finite conformal time. Every curvature invariant of $\hat{g}$ vanishes identically. The physical scalar curvature is $R = 12H^2$ and the cosmological constant $\lambda = 3H^2$. Nothing below depends on the exact de Sitter form beyond the asymptotics $a \sim e^{Ht}$, $\Omega \sim -H\eta$, which hold for any asymptotically de Sitter FLRW background with $H \to H_\infty > 0$; we keep H constant for clarity.

Definition 1 (Regular extension)**.** A field Φ on the physical spacetime, with conformal weight w, is said to extend *continuously* (resp. *smoothly*) to $\mathscr{I}^+$ if the rescaled field $\hat{\Phi} = \Omega^w \Phi$ admits a continuous (resp. C^∞) extension to the hypersurface $\Omega = 0$ in the unphysical spacetime.

In CCC the rescaled fields are required to be smooth across the crossover surface, so that the field equations of the next aeon can be posed there [2, 3]. Smoothness implies continuity; we shall find that for the Dirac field even continuity fails.

3 The scalar field: Friedrich's condition reread

Let φ be a real scalar field with action $-\frac{1}{2}\int\sqrt{-g}\,(g^{ab}\partial_a\varphi\partial_b\varphi + m^2\varphi^2 + \xi R\varphi^2)$, hence field equation $(\Box - \xi R - m^2)\varphi = 0$. Under $\hat{g} = \Omega^2 g$, the conformally coupled operator transforms covariantly:

$$\Big(\Box - \tfrac{1}{6}R\Big)\varphi = \Omega^3\Big(\hat{\Box} - \tfrac{1}{6}\hat{R}\Big)\hat{\varphi}, \qquad \hat{\varphi} = \Omega^{-1}\varphi. \tag{2}$$

Writing $\xi R + m^2 = \frac{1}{6}R + (m^2 + (\xi - \frac{1}{6})R)$, the field equation becomes, in the unphysical frame,

$$\Big(\hat{\Box} - \tfrac{1}{6}\hat{R}\Big)\hat{\varphi} = \frac{m^2 + (\xi - \frac{1}{6})R}{\Omega^2}\,\hat{\varphi}. \tag{3}$$

On de Sitter space, $R = 12H^2$, and the singular coefficient is $\big(m^2 - 2H^2(1 - 6\xi)\big)/\Omega^2$. For the minimally coupled field ($\xi = 0$) it vanishes if and only if

$$m^2 = 2H^2 = \frac{2\lambda}{3} = \frac{R}{6}, \tag{4}$$

which is Friedrich's condition [4]. The rereading is that the admissible mass is precisely the one for which $(\Box - m^2)\varphi = 0$ coincides with $(\Box - \frac{1}{6}R)\varphi = 0$: the only massive scalar that crosses $\mathscr{I}^+$ smoothly is the one that, on the background, is indistinguishable from the conformally coupled massless scalar. The compensation is possible because m^2 and R have the same dimension, so that a dimensionless coupling ξ can trade one for the other.

The homogeneous mode makes the nature of the obstruction explicit. With $\chi = a\varphi = \hat{\varphi}$ and $\xi = 0$, equation (3) at zero spatial momentum reads $\chi'' + \big(m^2/H^2 - 2\big)\eta^{-2}\chi = 0$, whose solutions are $\chi \propto |\eta|^s$ with

$$s = \tfrac{1}{2} \pm \mu, \qquad \mu = \sqrt{\tfrac{9}{4} - \tfrac{m^2}{H^2}}. \tag{5}$$

The exponents are the integers 0 and 1 only for $m^2 = 2H^2$. For $m^2 < 2H^2$ one branch diverges; for $2H^2 < m^2 \le \frac{9}{4}H^2$ both branches tend to zero as a non-integer power; for $m^2 > \frac{9}{4}H^2$ both tend to zero as $|\eta|^{1/2}$ times an oscillating phase $|\eta|^{\pm \mathrm{i}\sqrt{m^2/H^2 - 9/4}}$. In the physically relevant regime $m \gg H$, then, the rescaled scalar field *does* have a limit at $\mathscr{I}^+$, namely zero; what fails is differentiability. This is the sense in which Friedrich's obstruction is an obstruction to smoothness. The point matters for the comparison with the spinor.

4 The Dirac field

4.1 Conformal frame

Let ψ be a Dirac field of mass m, $(\mathrm{i}\nabla\!\!\!/ - m)\psi = 0$, on a spacetime (M, g). In four dimensions the massless Dirac operator is conformally covariant with weight 3/2: under $\hat{g} = \Omega^2 g$,

$$\hat{\nabla\!\!\!/}\,\hat{\psi} = \Omega^{-5/2}\,\nabla\!\!\!/\psi, \qquad \hat{\psi} = \Omega^{-3/2}\psi, \tag{6}$$

see e.g. [5, 6]. Hence the massive equation becomes, in the unphysical frame,

$$\Big(\mathrm{i}\hat{\nabla\!\!\!/} - \frac{m}{\Omega}\Big)\hat{\psi} = 0. \tag{7}$$

The mass appears as m/Ω, a simple pole on $\mathscr{I}^+$. The ratio of the mass to any frequency is conformally invariant, so a statement often made in the CCC literature, that masses become negligible near $\mathscr{I}^+$ because everything is redshifted, is not available: in the unphysical frame the mass dominates every other scale as $\Omega \to 0$.

4.2 No curvature coupling can cancel the pole

For the scalar, the pole m^2/Ω^2 could be cancelled by a dimensionless curvature coupling because R has the dimension of m^2. For the spinor, a term that could cancel m/Ω in (7) would have to be a Lorentz scalar (or a term of the same spinor structure as a mass) of dimension $(\text{length})^{-1}$, local in the curvature and its derivatives, and transforming under Weyl rescaling with weight -1. Every local scalar polynomial in the curvature and its derivatives has dimension $(\text{length})^{-2n}$ with $n \geq 1$; it can enter a mass-like term only with an explicit coefficient of dimension $(\text{length})^{2n-1}$, that is, with a new scale ℓ. Consider the most general such term at lowest order, $\ell R\psi$. On a constant-curvature background it is a shift of the mass, $m \mapsto m + \ell R$, and the pole in (7) becomes $(m + \ell R)/\Omega$. It vanishes only if $m + \ell R = 0$, i.e. if the *physical* effective mass vanishes on the background. There is no spinor analogue of the scalar situation, in which the physical field is massive (with Compton wavelength of order the Hubble length) and nonetheless crosses $\mathscr{I}^+$. For the Dirac field, regularity of the crossover is equivalent to the vanishing of the physical effective mass at $\mathscr{I}^+$; curvature couplings only relabel what "the mass" means.

4.3 The rescaled field has no limit at $\mathscr{I}^+$

We now take the de Sitter background of Section 2, in which $\hat{g}$ is flat and (7) reads, in Cartesian coordinates and with $a = 1/\Omega$,

$$\mathrm{i}\gamma^\mu \partial_\mu \hat{\psi} = m\, a(\eta)\, \hat{\psi}, \qquad a(\eta) = -\frac{1}{H\eta}. \tag{8}$$

For a spatial Fourier mode $\hat{\psi} = e^{\mathrm{i}\mathbf{k}\cdot\mathbf{x}} u(\eta)$ this is the Schrödinger-type equation

$$\mathrm{i}\,\partial_\eta u = \mathcal{H}(\eta)\, u, \qquad \mathcal{H}(\eta) = \boldsymbol{\alpha}\cdot\mathbf{k} + \beta\, m\, a(\eta), \tag{9}$$

with $\beta = \gamma^0$ and $\boldsymbol{\alpha} = \gamma^0\boldsymbol{\gamma}$ Hermitian. Two facts follow immediately.

Lemma 1 (Constant norm)**.** *For every mode, $\|u(\eta)\|^2 = u^\dagger u$ is independent of η. In particular no solution of (8) tends to zero at $\mathscr{I}^+$, and none diverges.*

Proof. $\mathcal{H}(\eta)$ is Hermitian for each η, so the evolution operator is unitary. (This is the conservation of the Dirac current $\hat{j}^0 = \hat{\psi}^\dagger\hat{\psi}$ in the flat unphysical frame; in the physical frame it is the statement that $a^3\psi^\dagger\psi$ is conserved for a homogeneous mode.) □

Proposition 1 (No limit at $\mathscr{I}^+$)**.** *Let $m \neq 0$. For every $\mathbf{k}$ and every non-zero solution u of (9), the limit $\lim_{\eta\to 0^-} u(\eta)$ does not exist. More precisely, there is a non-zero constant spinor χ_0, depending on the solution, such that*

$$u(\eta) = \exp\Big(\mathrm{i}\,\beta\,\frac{m}{H}\ln\Omega\Big)\,\chi(\eta), \qquad \chi(\eta) \to \chi_0 \ \text{as}\ \eta \to 0^-, \tag{10}$$

with $\|\chi(\eta) - \chi_0\| \leq k|\eta|\,\|\chi_0\|$, where $\Omega = -H\eta$ and $k = |\mathbf{k}|$.

Proof. For $\mathbf{k} = 0$ the system (9) is diagonal in the eigenbasis of β: on the eigenspaces $\beta = \pm 1$, $\mathrm{i}\partial_\eta u_\pm = \pm m a\, u_\pm$, and since $\int m\, a\, \mathrm{d}\eta = -(m/H)\ln|\eta| + \text{const} = (m/H)\ln a$,

$$u_\pm(\eta) = \exp\Big(\mp\mathrm{i}\frac{m}{H}\ln a\Big)u_\pm(\eta_0) = \exp\Big(\pm\mathrm{i}\frac{m}{H}\ln\Omega\Big)u_\pm(\eta_0) \qquad \text{(up to a constant phase)}, \tag{11}$$

which is (10) with χ constant. In physical time this is just $e^{\mp\mathrm{i}mt}$: the rescaling removes the decay of the amplitude but cannot remove the Compton oscillation, and $t \to \infty$ corresponds to $\ln\Omega \to -\infty$ at finite η.

For $\mathbf{k} \neq 0$, pass to the interaction picture with respect to the mass term: set $u(\eta) = U_0(\eta)\chi(\eta)$ with $U_0(\eta) = \exp(\mathrm{i}\beta\frac{m}{H}\ln\Omega)$, which is the exact $\mathbf{k} = 0$ evolution. Then $\mathrm{i}\partial_\eta\chi = U_0^\dagger(\boldsymbol{\alpha}\cdot\mathbf{k})U_0\,\chi =: V(\eta)\chi$, where $V(\eta)$ is Hermitian with $\|V(\eta)\| = k$ for all η. The Dyson series for the evolution of χ between η and η' converges in norm, and $\|\chi(\eta') - \chi(\eta)\| \leq k|\eta' - \eta|\,\|\chi\|$. Since $\int_\eta^0 \|V\|\,\mathrm{d}\eta' = k|\eta| < \infty$, $\chi(\eta)$ is Cauchy as $\eta \to 0^-$ and converges to some χ_0 with $\|\chi_0\| = \|u\| \neq 0$, which gives (10) and the bound.

Finally, decompose $\chi_0 = \chi_0^+ + \chi_0^-$ along the eigenspaces of β; at least one component is non-zero, say χ_0^+. The corresponding component of $u(\eta)$ is $e^{\mathrm{i}(m/H)\ln\Omega}\chi^+(\eta)$ with $\chi^+(\eta) \to \chi_0^+ \neq 0$. Along the sequence η_n defined by $(m/H)\ln\Omega(\eta_n) = -2\pi n$ it tends to χ_0^+, and along $(m/H)\ln\Omega(\eta_n') = -2\pi n - \pi$ it tends to $-\chi_0^+$. Hence u has no limit. □

Remark 1. *The modulus of u is constant and bounded, so the rescaled field is* bounded *at $\mathscr{I}^+$; what fails is the existence of a limit. The time derivative diverges: $\|\partial_\eta u\| \geq ma\|u\| - k\|u\| \sim m\|u\|/(H|\eta|)$. The structure (10) is the statement that the Dirac field on de Sitter belongs to the principal series for every $m \neq 0$: in the language of boundary conformal weights, $\Delta = \frac{3}{2} \pm \mathrm{i}m/H$, complex for all real $m \neq 0$, whereas the scalar has $\Delta = \frac{3}{2} \pm \mu$, real (complementary series) for $m^2 \leq \frac{9}{4}H^2$ and integer precisely at Friedrich's value. The same conclusion is reached from the exact mode functions, which are Hankel functions of order $\frac{1}{2} \mp \mathrm{i}m/H$ and behave as $|\eta|^{\pm\mathrm{i}m/H}$ at small argument.*

Remark 2 (Independence of the state)**.** *Proposition 1 is a statement about every solution of the classical field equation, hence about every mode function of the quantum field in every quasi-free state: the two-point function of a state in which the one-particle component with momentum $\mathbf{k}$ is occupied inherits the phase (10). A single electron present in the state suffices; the obstruction is not a statement about a density or a stress tensor.*

4.4 Orders of magnitude

For the electron, $m_e \simeq 5.11 \times 10^5\,\mathrm{eV}$, and for the asymptotic Hubble rate of our universe, $H_\infty = H_0\sqrt{\Omega_\Lambda} \simeq 1.2 \times 10^{-33}\,\mathrm{eV}$ (with $H_0 \simeq 67\ \mathrm{km\,s^{-1}Mpc^{-1}}$ and $\Omega_\Lambda \simeq 0.69$),

$$\frac{m_e}{H_\infty} \simeq 4 \times 10^{38}. \tag{12}$$

The phase of the rescaled electron field turns by about 4×10^{38} radians each time Ω decreases by a factor e, that is, during each Hubble time of the late universe. Conversely, for the rescaled field to turn by less than one radian between some late time and $\mathscr{I}^+$, the electron mass would have to be smaller than H_∞ over that whole interval. Any "fading" of rest mass that is to make the crossover continuous must therefore bring m_e below $10^{-33}\,\mathrm{eV}$, not merely make it small compared with some other scale.

5 Consequence for Conformal Cyclic Cosmology

CCC requires the rescaled matter fields of the earlier aeon to be smooth, or at least continuous, on the crossover surface, so that they can serve as data for the later aeon [2, 3]. Combining Sections 4.2 and 4:

Proposition 2. *Let the late-time geometry of an aeon be asymptotically de Sitter with $H \to H_\infty > 0$, and let the matter content include a Dirac field of constant non-zero mass m with at least one quantum present in the state near $\mathscr{I}^+$. Then the rescaled Dirac field has no continuous extension to $\mathscr{I}^+$, and no local curvature coupling of the field restores one without setting the physical effective mass to zero on the background. A conformal crossover that is continuous for that field requires $m \to 0$ at $\mathscr{I}^+$.*

Penrose's postulate that rest mass fades near the crossover [1] is therefore not an optional simplification: without it, the crossover fails for Dirac fields at the level of continuity, before any question of smoothness or of the conformal field equations arises. The postulate is also quantitatively demanding, as the previous section shows. In the Standard Model the electron is stable, its mass is a fixed parameter of the Higgs sector, and there is no process by which it vanishes at late times; Friedrich made the corresponding observation for the scalar [4]. In that precise sense, CCC with Standard-Model matter does not possess the crossover it postulates. We do not conclude that CCC is refuted, because the postulate of mass fading is part of the theory as Penrose states it; we conclude that the theory requires physics beyond the Standard Model at the crossover, and that this requirement is sharper for spinors than the scalar case suggests.

Two kinds of escape are logically open, and we name them without endorsing either. The first is that no electron, and more generally no massive fermion, is present in the state near $\mathscr{I}^+$: all matter would have to have been absorbed into black holes that subsequently evaporate into massless radiation, with no residue. Penrose himself has noted that isolated electrons generically remain [1], and the electron, as the lightest charged particle, cannot decay while charge is conserved. The second is a mechanism, not merely a postulate, by which the masses of all Dirac fields vanish in a cold, empty, asymptotically de Sitter universe. Since fermion masses arise from the Higgs vacuum expectation value, this amounts to a restoration of the electroweak symmetry at late times; the only condition under which the Higgs field itself crosses $\mathscr{I}^+$ regularly is Friedrich's, $m_h^2 = 2H_\infty^2$, which differs from the observed Higgs mass parameter in sign and by some 87 orders of magnitude. Whether a conformally invariant extension of the Standard Model (e.g. [7]) can provide such a mechanism is a question we leave open.

6 What is and is not established

- *Established here*: Proposition 1 (test Dirac field on de Sitter, every mode, every $m \neq 0$: constant norm, no limit at $\mathscr{I}^+$), Section 4.2 (no dimensionless curvature coupling can cancel m/Ω), and the rereading of Friedrich's condition as the conformal-coupling point (Section 3).

- *Assumed*: an asymptotically de Sitter late-time geometry with $H_\infty > 0$, which is what makes $\mathscr{I}^+$ spacelike and at finite conformal distance. For a dark-energy model in which $H \to 0$ the boundary is not of this type and the argument does not apply. The argument also treats the Dirac field as a test field on the background; this is weaker than Friedrich's coupled analysis, but sufficient for a negative result, since the test field is the zeroth order of any coupled system.

- *Not established*: that Dirac quanta are in fact present near $\mathscr{I}^+$ (the argument needs one); that the obstruction takes any particular form in the rescaled stress tensor or in the conformal field equations (we have deliberately stated it on the field, which is the weaker and safer statement); anything about the Big Bang side of the crossover, where the relevant conformal factor diverges rather than vanishing and the analysis is different.

- *Known to the literature*: the conformal covariance of the massless Dirac operator, the exact de Sitter mode functions, and the complex boundary weights of the de Sitter fermion. We have not found a statement in the CCC literature that the spinor obstruction is to continuity rather than smoothness, nor one that it cannot be removed by a curvature coupling; if such a statement exists we would be grateful to be told.

Verification

A short script, `dirac_scri_check.py` (Python, SciPy), integrates (9) numerically from $\eta = -1$ to $\eta = -10^{-6}$ for $(m/H, k/H) = (1, 0)$, $(1, 0.5)$ and $(3, 2)$. It confirms the conservation of the norm to 10^{-9}, the phase $-(m/H)\ln a$ of the $\mathbf{k} = 0$ solution to 10^{-10} modulo 2π, the convergence of the interaction-picture spinor χ (variation below $k|\eta|$), and the winding of the phase by $(m/H)\ln(10^6)$ radians over the interval. It also prints the scalar exponents of Section 3. The script and its output accompany this paper.

Provenance

This paper rewrites, in standard form and for an outside reader, a result obtained within an independent research programme on the conformal crossover of CCC, whose working records are deposited at Zenodo (DOI 10.5281/zenodo.23373604, version 2; concept DOI 10.5281/zenodo.22919750) and at `github.com/droppy94/LC_Raccord`. The derivation of Section 4 was first obtained in a working session of 23 September 2026 and is recorded there as S26-02, §2.3; the present paper adds the interaction-picture proof for $\mathbf{k} \neq 0$, the norm lemma and the argument of Section 4.2. The work was carried out with the assistance of an AI language model (Claude, Anthropic) under a pre-registration protocol; no human third party has reviewed it, which is the purpose of making it public. Comments and refutations are welcome.

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